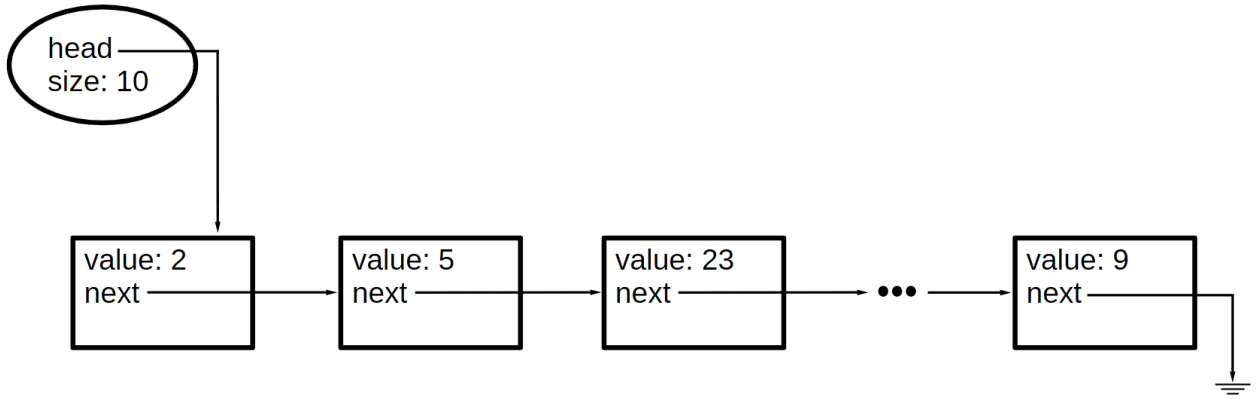
Linked List Implementation & Application

Task 1: Linked List implementation

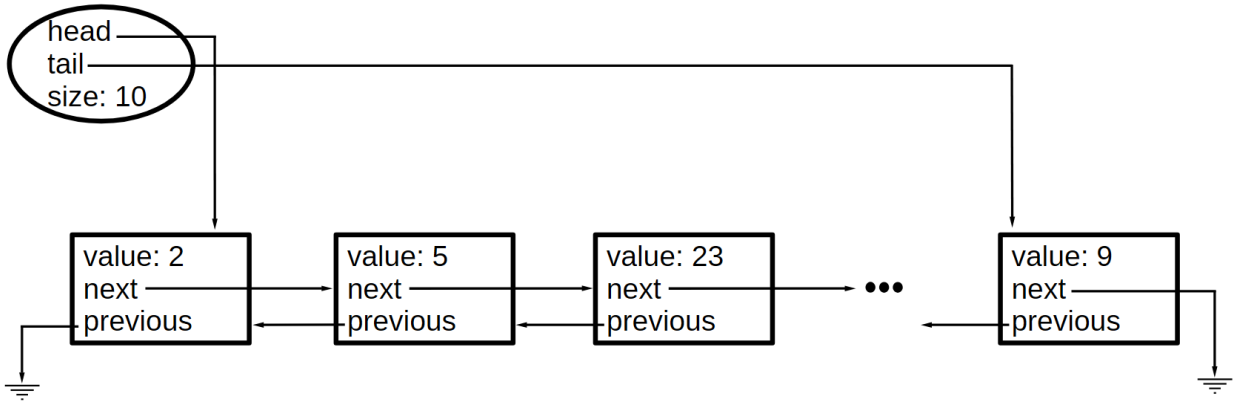
Linked lists are an essential data structure for computer systems. They complement dynamic arrays to overcome their limitation of needing a contiguous memory space to store their data. Linked lists eliminate that need by storing data anywhere memory is available, and then connecting those data together by having pointers pointing to the next memory location data is available. Real world applications of linked lists include process scheduling, memory management, program navigation like undo/redo, forward/backward buttons in enterprise applications, browsers, etc.

There are three main implementation of a linked lists, Singly Linked Lists, Doubly Linked Lists, and Circular Linked Lists as shown below.

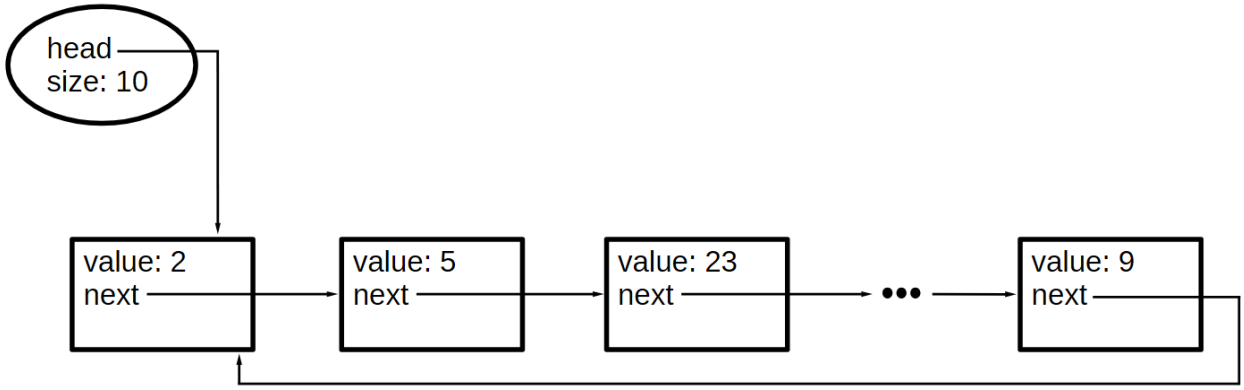
Singly Linked List



Doubly Linked Lists



Circular Linked Lists



For this task you need to implement a Doubly Linked List. Use the linkedList.c to implement the linked list functions mentioned in the linkedList.h as prototypes. You can use helper functions if you want, but make sure that the user has access to only what is necessary. Make sure your implementation includes all the safety precautions and that does not have any memory leaks!

Task 2: Deque implementation

Use the linkedList.c to implement the Deque all functions mentioned in the linkedList.h as prototypes. Note that the linked list is the data structure to use for the deque, therefore linked list functions may be the used for the deque implementation. Make sure your implementation includes all the safety precautions and that does not have any memory leaks!

Part 3: Test cases

Use the linkedListMain.c to write test cases for your linked list and deque implementations. Make sure that you test all the function mentioned in the linkedList.h file. Make sure that your functions are tests for both successful executions as well as executions that should fail.

Grading (100 pts) :

- 18 functions – 4 pts each
- Style & Documentation – 28 pts

What to submit:

- One archive containing linkedList.h, linkedList.c, linkedListMain.c and your compile script.

Extra Credit:

Once you're done and have submitted everything, there's one last thing that takes 5-10 minutes! As Silas announced in class, he is running research on how students use AI. Please use this link below to either complete the research study or do the alternative reflection activity. Participation in the research is voluntary and confidential to the course instructors, but you must either complete the survey or the reflection activity and explicitly decline participation to receive credit.

https://stevens.qualtrics.com/jfe/form/SV_42w3fWc8opmT0l8?Course=CSC2050-07&Assignment=3